

Table 1: List of cognitive or social biases with impact on user-centric (X)AI interactions

Bias	Description
Anchoring bias	Tendency to rely on the first piece of information encountered (the “anchor”) when making decisions
Anthropomorphism bias	Attribution of human-like characteristics to other living organisms (like animals) or non-living, abstract objects (like algorithms, agents)
Authority bias	People’s tendency to follow commands uncritically. Usually in hierarchical interactions (co-pilot vs. captain) and/or in collective cultural groups.
Automation bias	Over-relying on automated systems and favoring suggestions from these systems, even when they might be incorrect
Confirmation bias	Focusing on information that confirms one’s preconceptions, ignoring contradictory information
Expectation bias	Having preconceived notions about the capabilities or intentions of an artificial agent based on its design, appearance, or prior experiences
Negativity bias	Giving more weight to negative experiences with an artificial agent than positive ones, potentially leading to quicker dissatisfaction or mistrust
Observer-Expectancy bias	Unconscious influence the participants of an experiment by the experimenter (relevant item in the experiment protocol). Links to “Social Facilitation/Inhibition”: Performing better or worse on tasks when being observed by or interacting with an artificial agent, similar to the effect of human observers